Guide for Model Administrators

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SAP Business Network Global Track and Trace | LBN

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Introduction

About This Document

This documentation describes how to perform all the tasks required to manage version 2 models for SAP Business Network Global Track and Trace. This involves using the following apps:

- Manage Models (MM) app
- Check Processes and Events (CPE) app

Target Audience

The target audience for this guide is Administrator – Data Models (GTT).

Prerequisites

You are able to sign in to SAP Business Network Global Track and Trace as an Administrator – Data Models (GTT).

About Global Track and Trace

The main purpose of SAP Business Network Global Track and Trace is to track processes that are influenced by business events. All tracked processes and business events have a common core behavior and share a common set of basic attributes.

Therefore, SAP Business Network Global Track and Trace consists of the following:

- A core engine: provides generic functionality to track processes, handle business events and provide the modeling environment described in this document.
- GTT apps: enrich the core engine's generic functionality by adding domain specific semantics in terms of data and behavior.

Basic Terms

The applications of SAP Business Network Global Track and Trace deal with the following central objects and terms:

Tracked Process

A tracked process comprises the following:

- Involved parties acting in or being affected by the process.
- Track multiple other processes.
- A set of planned events that are expected to happen within the tracked process.

> **Note**
>
> Planned events are called "milestones" in SAP ERP applications like SAP Event Management.

- A set of events that actually happened against the process. This could be a realization of a planned event or also an unplanned event.

Event

Events can be planned events or unplanned events that might occur during a process. When an event happens, it becomes an actual event. An actual event needs to be matched against a planned event of that process. If not, it becomes an unplanned event.

Planned Event

A planned event:

- Has a fixed semantic.
- Has a planned date and time (with time zone), a tolerance window, a planned location, and other planned data.

Examples include:

- goods issued (planned date: 2016-07-06; planned location: gate 1)
- goods received (planned date: 2016-07-07; planned location: pack station 801)

Unplanned Event

A unplanned event:

- Has a fixed semantic as well.
- Is not planned, but can be foreseen that it occurs with a certain likelihood, but cannot be planned thus has no planned data.
- Examples include: delay in delivery, damage of goods.
- Or is a piece of information at an arbitrary point in time, such as the current position and temperature of a container.

Actual Event

An actual event:

- Is an event that actually happened.
- Is either matched against a planned event of a process and complements planned data with actual data or becomes an unplanned event of the process.
- Triggers changes of attributes of a tracked process.

Timestamps

Every event, planned or actual, has two timestamps:

1. (planned/actual) business timestamp: the time at which the event physically happens.
2. (planned/actual) technical timestamp: the time at which a user reports the event in the system.

Usually, the timestamps for the same event are different, with the time of the business timestamp before that of the technical.

The timestamp uses the UTC time.

The event status is determined by comparing the planned business timestamp with that of the actual.

ℹ Note

For simplicity in the following examples, only one planned business timestamp is used to determine the event status. In practice, rather than a single planned business timestamp, a range is used from the planned business earliest timestamp to the planned business latest timestamp.

Example 1:

The times in the timestamps for an “arrival” event are as follows:

- Planned business timestamp: 2:00 pm
- Actual business timestamp: 2:30 pm
- Actual technical timestamp: 3:00 pm

The comparison of planned and actual business timestamps determines the event status:

- Planned business timestamp: 2:00 pm
- Actual business timestamp: 2:30 pm
- So the event status is **Late**.

Reporting an Event Again:

The situation is more complicated if the same event is reported again, or even multiple times. Then the sequence of actual events is determined by the setting you made when you created or edited the tracked process for **Affect Planned Event Status By**. You can select either: **Actual Business Timestamp** or **Actual Technical Timestamp**. For more information, see Create Tracked Process.

But in either case, the new timestamp must be later than the previous one, otherwise it is ignored.

Example 2:

As in the previous example, the times in the timestamps for an “arrival” event are as follows:

- Planned business timestamp: 2:00 pm
- Actual business timestamp 1: 2:30 pm (So the event status is Late.)
- Actual technical timestamp 1: 3:00 pm

The event is reported for a second time with the following timestamps:

- Actual business timestamp 2: 1:00 pm
- Actual technical timestamp 2: 4:00 pm

If you set: **Affect Planned Event Status By** actual business timestamp, then:

- The actual business timestamp is used to determine the sequence of actual events.
- As the second actual business timestamp (1:00 pm) is before that of the first (2:30 pm), the second timestamp is ignored.

- So, the actual business timestamp remains as 2:30 pm and the event status remains **Late**.

If you set: **Affect Planned Event Status By** actual technical timestamp, then:

- The actual technical timestamp is used to determine the sequence of actual events.
- As the second actual technical timestamp (4:00 pm) is after that of the first (3:00 pm), the second timestamp is valid and the actual technical timestamp becomes 4:00 pm.
- Event status is then determined by comparing the second event's actual business timestamp with that of the planned.

As the actual business timestamp (1:00 pm) is before the planned business timestamp (2:00 pm), the event status becomes **Early**.

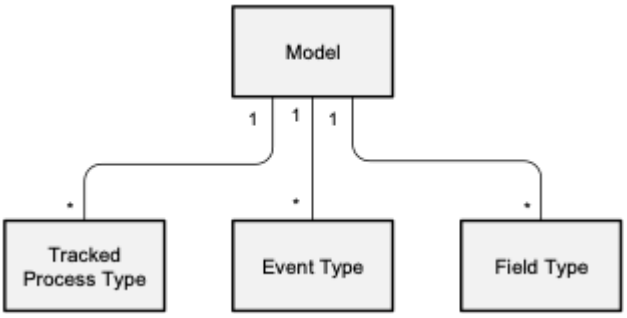
Structure

Different models can be created in the MM app. Each model must contain one or more tracked process types and event types. Each tracked process type and event type contain some predefined core model fields that you cannot change. You can add user-defined fields to each tracked process type and event type. For each tracked process type, some event types can be specified as an:

- **Admissible Planned Event** type: specifies the event types that can be used in planned events of a tracked process with this process type.
- **Admissible Unplanned Event** type: specifies the event types that can be used in unplanned events of a tracked process with this process type.

As shown in the following diagram, some other field types can be defined:

- Tracked process type
- Event type
- Field type



Model Overview Page

The MM app starts by opening the ***Model Overview*** page. This lists up to 20 models. The model count is displayed at the bottom. For example: (20/100) indicates that 20 models are displayed of a total of 100. If there are more than 20 models, click the More button to see the next 20. The information on each model is displayed on a model card.

To find your model, you can simply check the list or search by typing its model name. To select your model, click its model card.

Also, on the ***Model Overview*** page, you can do the following:

- Create a model
- Import a model
- Export a model
- Delete a model
- Refresh the latest data of models on the page by clicking the Refresh icon.

On the top right, a **Sample Codes** menu is provided with a direct link to download the sample codes:

- **For Fulfillment Tracking Apps**: links to the following GTT repository in the SAP Samples organization on GitHub.com where you can download the SAP ERP sample codes of fulfillment tracking apps: https://github.com/SAP-samples/logistics-business-network-gtt-standardapps-samples

Help and Support

There are several ways you can find help and support on the Fiori launchpad of SAP Business Network Global Track and Trace.

Built-In Support

Built-in support provides efficient, simplified and guided support channels, you can search for knowledge, ask for needed information, get help from recommended solutions. Currently, expert chat is not supported.

To access the built-in support service on the Fiori Launchpad, click the icon on the right corner of the screen. You will see a welcome page where you can sign in, and a list of recommended links for the page.

To gain full access to the built-in functions, you must activate your S-user. For more, see Activating Built-In Support Services.

For more information on how to use this tool, refer to Built-In Support on SAP Help Portal.

In-app Help

When you launch the app, you can turn on the in-app help that provides on-screen explanations of key fields and areas on the screen.

To turn on the in-app help, on the top right on the screen, click the icon.

Once the in-app help is turned on, you can:

- search the displayed help topics for the text that you type;
- click the icon to see the help texts on that topic;
- click the link tiles to see the documentation of this app on SAP Help Portal;
- hide the **Help Topics** panel by clicking the » button on the bottom right.

Standard and User Models

You have a choice of two types of model:

1. Standard models are predefined and delivered in the product.

2. User models are created by model administrators.

The standard model is consumed by the Fulfillment Tracking apps. It provides:

- standard tracked processes with model fields, planned and unplanned events in the **Tracked Process** tab
- standard field types with model fields in the **Field Type Pool** tab
- standard event types with model fields in the **Event Type Pool** tab
- standard code lists with model codes in the **Code List** tab
- standard IDOC mapping in the **IDOC Integration** tab
- standard visibility provider mapping in the **Visibility Provider Integration** tab
- standard planned event extensions in the **Planned Event Extension** tab
- standard declaration and script in the **Event to Action** tab.

You cannot delete a standard model. You can edit it to create user-defined tracked processes, field types, event types, code lists, planned event extensions, and event-to-action scripts. User-defined fields in the standard model also support IDOC mapping and visibility provider mapping.

ℹ Note

The name of user-defined tracked processes, field types, event types, code lists, and planned event extensions must start with the case-insensitive letters “ZZ”.

The standard model is integrated with LBN code lists.

The Model Card

The information displayed on the model card includes:

- **Name**: in lower case with numerals except as the first character. For example, process17 or salesorder317. The name must be unique in the tenant.
- Draft version status:
  - Blank: the model’s draft version is the same as the deployed version.
  - Draft: displayed as 

. The model’s draft version is different from the deployed version.
- Deployed version status:
  - Blank: the model has not been deployed.
  - Active: displayed as 

. The deployed version is active, meaning that the write/read services of the model are running.
  - Inactive: displayed as 

. The deployed version is inactive, meaning that the write/read services of the model are disabled.
- **Description**: up to 255 characters describing the model.
- **Namespace**: identifies the model and includes its name. This is unique across SAP Business Network for Logistics.

- **Last Operation Status**: The icon for each stage is color-coded, so you can see its status at a glance: green for success, red for error, and orange for warning and pending.
    - **Deployment Success**
    - **Deployment Warning**
    - **Deployment Pending**
    - **Deployment Error**
    - **Deletion Pending**
    - **Deletion Error**
- **Event to Action Log**: For more, see Enable/Disable Event to Action Log of the Model.
    - **Off**: the Event to Action Log is disabled.
    - **On**: the Event to Action Log is enabled.
- **Email Rules**: takes you to the **Email Rules** page where you can create and define email rules for email notifications. For more, see Define the Rule on the Configuration UI

Import Model

Context

You can import a single model from a JSON file that is sized below 2MB.

Procedure

1. On the **Model Overview** page, click the **Import** button
2. In the pop-up dialog, click the **Browse...** button to select a JSON file to upload. Once you have selected your file, click **Import**.
3. When the file is uploaded, you see one of the following three kinds of messages:
    - **Error**: The message identifies what went wrong, such as a broken file, an unreadable file or one with incompatible data structure. It may also suggest what to do next.
    - **Warning**: When the name of the uploaded model already exists in the MM app, you must confirm whether to overwrite the existing model or abort the upload.
    - **Success**: Your imported model was created successfully.

Export Model

Context

You can export a single model as a JSON file.

Procedure

1. On the **Model Overview** page, select the model you want to export, then click the **Export** button.

2. A pop-up dialog appears. If the model is already deployed, you can choose to export either the draft version or the deployed version. If the model is not deployed, you can only export the draft version.
3. After you have chosen the version you want, click **Export**. The exported JSON file is automatically downloaded to your browser. You can find it in your local download file path

Delete Model

Context

You can delete one model at a time.

Procedure

1. On the **Model Overview** page, select the model you want to delete, then click the **Delete** button.
2. A confirmation message box appears. You can choose to continue or abort.
3. If you continue, you get a success message when deletion is complete. If you meet an error, the **Last Operation Status** becomes **Deletion Error**, and you can view the model history to check the details.

Related Information

View the Model History

Create and Deploy a Model

Create a Model from Scratch

Context

The MM app allows you to create a model from scratch.

Procedure

1. Launch the MM app.
2. On the **Model Overview** page, click the **Create** button. The detail page of a blank model then appears.
3. On the **Model Details** page, you can do the following:
    - In the **Header** tab, fill in the model header information. See Model Header for details.
    - In the **Tracked Process** tab, create one or more tracked processes. In each tracked process, create user-defined fields, planned events and unplanned events. See Tracked Process for details.
    - In the **Field Type Pool** tab, create field types. They can be used as the **Composition** type when you define user-defined fields for a tracked process. See Field Type for details.
    - In the **Event Type Pool** tab, create event types. They can be used when you define planned/unplanned events for a tracked process. See Event Type for details.
    - In the **Code List** tab, create code lists. They can be used as the **Code List** type when you define user-defined fields for a tracked process. See Code List for details.

- In the **IDOC Integration** tab, define the IDOC integration mapping rules for a tracked process. See IDOC Integration for details.
- In the **Visibility Provider Integration** tab, define the visibility provider integration mapping rules for a tracked process. See Visibility Provider Integration for details.
- In the **Planned Event Extension** tab, create planned event extension fields. See Planned Event Extension for details.
- In the **Event to Action** tab, define the event-to-action rules for the model. (This replaces the business rules used in the Version 1 of SAP Business Network Global Track and Trace.) See Event to Action (Business Rules) for details.

4. Save your model configuration.

Edit an Existing Model

Context

The MM app allows you to import and edit a model.

Procedure

1. Launch the MM app.
2. On the **Model Overview** page, you can either:
   - import the model you want to edit by clicking the **Import** button,
   - or select an existing model.

   The detail page of a model then appears.
3. On the top right, click the **Edit** button.
4. On the **Model Details** page, you can do the following:
   - In the **Header** tab, edit the model header information. See Model Header for details.
   - In the **Tracked Process** tab, edit tracked processes or create new ones. In each tracked process, edit/create user-defined fields, planned events and unplanned events. See Tracked Process for details.
   - In the **Field Type Pool** tab, edit field types or create new ones. They can be used as the **Composition** type when you define user-defined fields for a tracked process. See Field Type for details.
   - In the **Event Type Pool** tab, edit event types or create new ones. They can be used when you define planned/unplanned events for a tracked process. See Event Type for details.
   - In the **Code List** tab, edit code lists or create new ones. They can be used as the **Code List** type when you define user-defined fields for a tracked process. See Code List for details.
   - In the **IDOC Integration** tab, edit the IDOC integration mapping rules for a tracked process or define new ones. See IDOC Integration for details.
   - In the **Visibility Provider Integration** tab, edit the visibility provider integration mapping rules for a tracked process or define new ones. See Visibility Provider Integration for details.
   - In the **Planned Event Extension** tab, edit planned event extension fields or create new ones. See Planned Event Extension for details.
   - In the **Event to Action** tab, edit the event-to-action rules for the model or define new ones. See Event to Action (Business Rules) for details.